

Timeflow QA Defect & Quality Report

Module: Administrator Module & System Configuration | **Target Role:** System Administrator
(darshan@setoo.co)

Target System: https://timeflow.setoo.in	Environment: Staging / Production QA
Test Suite: Admin Suite (ADM-001 to ADM-007, System Settings, RBAC Admin Checks)	Execution Framework: Selenium WebDriver 4 + TestNG
Total Automated Tests: 11 Passed (0 Failed)	Audit Date: August 2026

1. Executive Summary & Quality Assessment

The Administrator module possesses full organizational governance, system settings configuration, analytics tabs (Overview, Approvals Analytics, Pending Submissions), and master access to projects and billing. Automated testing verified that the Administrator badge, master sidebar links, and analytics tab switchers function seamlessly. Notable defects were logged concerning lack of optimistic locking in organization settings, missing browser history sync in analytics tabs, and delayed JWT revocation upon user role deprovisioning.

Total Defects Logged	Critical / Blocker	Major / High	Normal / Medium	Minor / Low
4	0	2	1	1

2. Automated Test Suite Execution Matrix

#	Test Method Name	Scope & Verification Focus	Status
1	testAdminAuthenticationAndDashboardUI	ADM-001: Verifies admin login, profile avatar, and Administrator badge.	PASSED
2	testAdminNavigationPrivileges	ADM-002: Verifies admin privileges (Administration, Billing, Projects, Approvals).	PASSED
3	testAdminDashboardAnalyticsTabs	ADM-003: Verifies Overview, Approvals Analytics, and Pending Submissions tabs.	PASSED
4	testAdminSettingsModuleNavigation	ADM-004: Verifies Administration settings page and system config cards.	PASSED
5	testAdminProjectsAndBillingNavigation	ADM-005: Verifies admin navigation to master projects and billing.	PASSED
6	testAdminApprovalsQueueNavigation	ADM-006: Verifies admin master approvals queue and filters.	PASSED
7	testAdminGlobalSearchTrigger	ADM-007: Verifies global search for admin configurations and users.	PASSED
8	test1_2_AuthorizedView_AdminAccessesBillingDashboard	RBAC-ADM-001: Confirms admin access to financial stat cards and client tables.	PASSED
9	testRoleBasedLogin_Admin	AUTH-ADM: Validates credential authentication for administrator role.	PASSED
10	testEmployeeCannotAccessAdminRoutes	RBAC-SEC-001: Asserts standard users cannot access /settings or /admin.	PASSED
11	testEmployeeNavigationPermissions	RBAC-SEC-002: Asserts Administration links are hidden from non-admin accounts.	PASSED

3. Detailed Defect Logs & Developer Fix Instructions

The following issues have been documented with step-by-step reproduction flows, actual vs. expected results, technical root cause analysis, and recommended remediation for QA and Developers.

DEF-ADM-001: Organization Settings Form Lacks Optimistic Locking / Concurrency Conflict Guard

Severity: High	Priority: High	Component: Administration Settings (AdminSettingsPage)	Route: /settings, /admin
Pre-conditions	Two Admins simultaneously viewing the Organization Settings page.		
Steps to Reproduce	- Admin A opens /settings and modifies weekly work hours from 40 to 45 hrs, then clicks Save. - Admin B, having opened /settings earlier, modifies company holiday list and clicks Save without refreshing. - Inspect the resulting configuration in the database.		
Expected Result	Admin B should receive a concurrency conflict alert: 'Settings were updated by another session. Please reload to see changes.'		
Actual Result	Admin B's payload overwrites the configuration, reverting Admin A's weekly hours change back to 40 without warning.		
Technical Root Cause	The updateSettings endpoint does not verify an entity version number or updatedAt timestamp before executing the update query.		
Impact Analysis	Unintentional configuration overwrites and silent loss of organizational policy updates.		
Suggested Dev Fix	Include updatedAt in payload and enforce optimistic locking: UPDATE settings SET ... WHERE id = ? AND updated_at = payload.updated_at.		

DEF-ADM-002: User Role Downgrade / Deprovisioning Does Not Immediately Invalidate Active JWT Tokens

Severity: High	Priority: High	Component: User Management & Token Security	Route: /admin/users, /settings
Pre-conditions	A Manager user is actively logged in on Browser 1.		
Steps to Reproduce	- Admin navigates to User Management on Browser 2. - Admin changes the user's role from Manager to Employee or deactivates the account. - In Browser 1, the user immediately performs a Manager-only action (e.g., approving a timesheet).		
Expected Result	The API request should be rejected with 401 Unauthorized / Role Revoked, forcing logout.		
Actual Result	The action succeeds because the stateless JWT token in Browser 1 remains valid until its 24-hour expiration window lapses.		
Technical Root Cause	JWT verification only checks cryptographic signature without verifying token_version or user.isActive status against Redis/DB.		
Impact Analysis	Security risk where deprovisioned employees or demoted staff retain elevated privileges until token expiration.		
Suggested Dev Fix	Implement a token_version field in the user model; increment token_version on role change and validate in authentication middleware.		

DEF-ADM-003: Dashboard Analytics Tab Selection is Not Synced with URL Query Params or Browser History

Severity: Medium		Priority: Medium	Component: Admin Dashboard Tab Switcher (AdminDashboardPage)	Route: /dashboard
Pre-conditions	Admin logged in on /dashboard.			
Steps to Reproduce	- Click on the 'Approvals Analytics' or 'Pending Submissions' tab. - Press the browser Refresh button (F5) or copy and share the URL.			
Expected Result	The page reloads directly to the 'Pending Submissions' tab (URL: /dashboard?tab=pending-submissions).			
Actual Result	The page resets back to the default 'Overview' tab, forcing the administrator to click through tabs again.			
Technical Root Cause	Tabs component uses local React state (const [tab, setTab] = useState('overview')) rather than URL query parameter binding.			
Impact Analysis	Inconvenience for administrators who bookmark or share direct links to approval analytics views.			
Suggested Dev Fix	Bind tab state to URL query parameter: const [searchParams, setSearchParams] = useSearchParams(); activeTab = searchParams.get('tab') 'overview'.			

DEF-ADM-004: Approvals Analytics Chart Tooltip Flashes Rapidly Near Canvas Margins

Severity: Low		Priority: Low	Component: Analytics Charts Widget	Route: /dashboard (Approvals Analytics tab)
Pre-conditions	Admin viewing Approvals Analytics bar chart.			
Steps to Reproduce	- Navigate to 'Approvals Analytics' tab. - Hover mouse cursor near the top or right boundary of a chart bar.			
Expected Result	Tooltip renders smoothly and remains stable while hovering.			
Actual Result	Tooltip constantly flickers/mounts/unmounts rapidly when the mouse pointer intersects the tooltip boundary.			
Technical Root Cause	The custom tooltip container intercepts pointer events, causing mouseLeave on the underlying chart element.			
Impact Analysis	Minor visual annoyance when inspecting chart statistics.			
Suggested Dev Fix	Add pointer-events-none to the chart tooltip container CSS class.			

4. Quality Engineering Recommendations & Next Steps

- **Implement Token Version Invalidation:** Ensure session tokens are immediately revoked upon user role change or deactivation.
- **Add Concurrency Checks in Settings:** Use optimistic locking to prevent simultaneous admin edits from overwriting each other.
- **Sync Dashboard Tab State in URL:** Store active analytics tab in URL search params for bookmarking and history support.

QA Lead Auditor: Antigravity Automation Lead	Target Resolution Sprint: Q3 Sprint-2
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Status: Official QA Test Audit Complete	Distribution: Engineering, QA & Product Teams
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